THE LANGUAGE OF BUSINESS ANALYTICS

01 FOUNDATIONAL CONCEPTS OF ANALYTICS

1. BUSINESS ANALYTICS: This is a process of using raw or unstructured information to make sound business decisions and to bolster business performance by reducing risk, increasing profit margins and lessening risks. Business Analytics uses patterns, trends, business knowledge and data visualisation tools to solve business challenges. Business Analytics is much broader than Data Analytics and Business Intelligence in that it focuses more on data within the business operations and using it to forecast future demand and achieve business goals with minimum cost.

Example: A supermarket analyses customer purchase data to decide which products to stock, changes in product prices, and demand forecast (future need for product)

Reference:

Evans, J.R (2017) Business Analytics. Pearson.
McKinsey & Company (n.d.) Analytics insights.
Meta Platforms, Inc. (2026) WhatsApp Meta AI [Large Language model]. Available at: WhatsApp (Accessed: 27 April 2026)

2. DATA ANALYTICS: This is a process of investigating raw data such as numbers, texts, customer feedback and images, and turning them into patterns and trends to find answers to what happened, why it happened, what could happen in the future and how the problem can be solved. This data needs to be cleaned and organised in order to get correct information.

Example: A data specialist queries a database to find or investigate changes in monthly revenue of a business.

The key difference about data analytics is that any data can be inspected, not only business data as opposed to business analytics which focuses on company / business data to extract insights on where the business is going.

Reference:

Investopedia (n.d.) Data Analytics

3. <u>BUSINESS INTELLIGENCE</u>: This gives company stakeholders the direction of the company, where the company is in terms of performance and the pace at which it is going. It involves the use of tools and technologies such as charts, bar graphs and reports to present information about the business, in the past and the current time.
The key difference is that with Business Analytics, it focuses on the future of the business and what remedies are required to move forward and improve business performance

Example: Just like a vehicle has a speedometer, fuel and oil gauges, and global positioning system, just so a business will have a data specialist that will create a dashboard using tools such as Power BI to illustrate daily sales, sales growth (monthly/quarterly), current expenditure.

Reference:

Investopedia (n.d.) Business Intelligence (BI)
Coursera (2026) BI vs Data Analytics: What's the Difference?

Available at: https://m.youtube.com/watch?v=8W-gRNMEGHc
(Accessed: 28 April 2026)

4. DATA-DRIVEN DECISION MAKING: This involves making business decisions by using all types of data (structured or unstructured) to forecast business performance. This requires data analysis not guesses or instinct. This is a business practice or culture rather than a tool or technology such as Artificial intelligence.

Example: Applying the culture if the historical data of a product being bought is there. i.e If a Gucci bag costs R50 000 and you have past sales data you can use it to predict future sales and demand but if you are launching it for the very first time it is intuition based.

Reference:

Harvard Business Review (n.d.) Data-driven decision making articles.

Provost, F. and Fawcett, T. (2013) Data Science for Business: What You Need To Know about Data Mining and Data-Analytic thinking. Sebastopol: O'Reilly Media.

5. BIG DATA: This refers to massive quantities of data sets that are complex and grow quickly to the extend that their capacity cannot be hosted by traditional data processing systems like Excel spreadsheets. in order to make effective decision making. This is just large amounts of raw data as opposed to analytics where this data is categorised, refined, and converted into meaningful information.

Example: The use of tools such as BigQuery to track millions of customer transactions weekly acaross various stores based in different locations (including weather) to predict what beverage to recommend to a client the following day

Reference:

McKinsey & Company (n.d.) Big data insights.
Statista (n.d.) Big data statistics.
IBM (2021) What is Big Data? Available at: https://www.ibm.com/think/topics/big-data (Accessed: 28 April 2026).

6. Descriptive Analytics: This answers the question of what happen d. Data from the past is added aggregated/ together to better understand previous performance. This does not predict the future nor explain the events causing performance.

Example: Sales were R300 000.00 in February 2026, lower than March 2026 by 12%. Cape Town Store made 37% of total revenue.

Reference:

Evans, J.R (2017) Business Analytics. Pearson.

7. Diagnostic Analytics: This refers to the inspecting of data establish the causes of historical results. In business terms it is referred as 'Root Cause Analysis' and requires to answer "What happened?" first.

Reference:

Evans, J.R. (2017) Business Analytics. Pearson.

Example: A drop in sales of cooking oil was caused by price increases.

8. Pridictive Analysis: This refers to an advanced form of using historical data to predict or forecast future events and results.

Example: Retailer stocking specific quantities for a certain month based on last year's sales and current weather condition

Reference:

McKinsey & Company (n.d.) Analytics & Big Data Insights

IBM (n.d.) Predictive Analytics. Available at: https://www.ibm.com/think/topics/predictive-analytics (Accessed: 28 April 2026)

9. PRESCRIPTIVE ANALYTICS: This refers to recommendations that are given after using using tools and decision-making models to answer what happened, why it happened and what could likely occur in the future. This analysis ensures provides contingency plans in case of ~~adv~~ unexpected events

Example: Mines will keep buffer stock of a fast moving commodity in case a supplier delivers late.

Reference:

McKinsey & Company (n.d) Analytics & Big Data Insights

10. Data Literacy: This refers to being able to assertively work with data whilst examining it and understanding it as well as transferring the knowledge effectively. This is rather a skill that one needs to acquire through training and repetitive learning using numbers as opposed to just words (qualitative analysis)

Example: When a customer survey suggests that 90% of customers prefer ice-cream to cold drink whilst only 20 customers were surveyed in the store. This is almost too perfect. Therefore an analyst will need to investigate the numbers.

Reference.
HubSpot Academy (n.d.) Data Literacy Training
Meta Platforms, Inc (2026) WhatsApp Meta AI [Large Language model]: Available at: WhatsApp (Accessed: 28 April 2026)
Saunders, M., Lewis, P. and Thornhill A. (2019) Research Methods for Business Students. 8th edn. Harlow: Pearson. p. 294-299.

02. DATA, MEASUREMENT & REPORTING

11. STRUCTURED DATA: This is data that is extremely organised, stored in a systematic manner in a predetermined format with rows and columns. The data is tabulated and it is reddy to be interpreted, stored, examined and queried. This can be sales transactions in a POS system.

Reference:
Evans, J.R (2017) Business Analytics
Investopedia (n.d.) Structured Data

12 UNSTRUCTURED DATA: This data does not have any structured format. It does not have any systematic way of storing the data. This makes it very complex to query, analyze and store. As result it will require tools that are more advanced to structure it. Data such customer reviewes and CCTV footage would require tools that are advanced such as Natural Language Processing (NLP)

in order to interpret it.
Reference:
Mickeisy & Company (n.d.) Analytics and benchmarking insights
Statista (n.d.) Industry benchmarks and data statistics.

13. DATA WAREHOUSE: central locating that stores and combines data from various databases and other sources for analysing and reporting purposes. A retail chain would merge data from Point-Of-Sale (POS) systems into a single warehouse for reporting.

Evans, J.R. (2017) Business Analytics. Pearson.
McKinsey & Company (n.d.) Analytics and benchmarking insights.

14. DATA MINING: This is a process of investing large datasets to detect patterns, trends and relationships in outcomes and their events. An analysis would show that a customer who buys bread would also ~~by~~ buy milk.
Reference
Investopedia (n.d.) Data Mining
Evans, J.R. (2017) Business Analytics. Pearson.

15. DATA V. UALIZATION: This is an illustrative manner of presenting information in the form of graphs, to make it simper to understand and relay to others. This can be a dashboard ~~consisting~~ illustrating of customer growth charts and profit margins.
Reference:
Harvard Business Review (n.d.) Da - driven decision making KPIs
HubSpot Academy (n.d.) Marketing Analytics & Data Literacy

16. DASHBOARD: This is a point of interaction between the users & and a system which depicts critical numeric data that is measurable in a live format in order to monitor performance and to respond to it timeously. Google Analytics is able to show conversion rates e.g. A customer visiting your website and completing a purchase on your e-commerce store.

Reference:

Graphed.com (2025) What is conversation rate in Google Analytics Available at: graphed.com (Accessed: 2 May 2026)
Harvard Business Review (n.d.) Data-driven decision and KPIs
Google Analytics Help (n.d.) Metrics, dashboards, and segmentation.

17. METRIC: Any measurable number used to trace or detect performance of an activity. This can be the number of customers or daily sales.

Reference:

Investopedia (2025) Understanding Metrics: Key to Performance Tracking and Analysis Available at: https://www.investopedia.com/terms/m/metrics.asp. (Accessed: 2 May 2026)

18. KEY PERFORMANCE INDICATOR (KPI): This is a critical and measurable quantity that shows how well is the business is performing in attaining its main objectives.
Every KPI is a metric (a quanitifyable number) but not every metric is a KPI (because every company has it own objective/strategy which decide which metrics should be focused on). For an e-commerce store conversion rate (visitors making a purchase) is a KPI but time on the site/or store is a metric. For a News Blog time on the site may be key (KPI) but conversion rate (signing up) may be a metric.

Reference:
HubSpot (n.d.) Key Performance Indicators Available at: www.hubspot.com/glossary/Key-performance-indicators (Accessed: 2 May 2026)

19. SEGMENTATION: A process of grouping customers according to similar attributes such as age, location and behaviour. A marketind department groups customers so that they can target each group according to their buying behaviour.
Ref
HubSpot Academy (n.d) Marketing Analytics & Data Literacy

20. BENCHMARK: A standard that is used compare one companies performance against industry norms. This can be comparing your restaurant's profit margin with industry averages.
Reference:
Statista

03 CUSTOMER & SALES KPIs

21 CONVERSION RATE: A percentage of users who fufills a company's key performance indicator. An e-commerce store can use Google Analytics to track how many visitors actually buy their products. Conversion Rate = $\frac{\text{Conversions}}{\text{Total visitors}} \times 100$
Reference:
Google Analytics Help (n.d.) Conversion tracking and metrics. Available at: https://support.google.com/analytics (Accessed: 2 May 2026)

22. CUSTOMER ACQUISITION COST: The total amount in promotions and related costs in order to gain a new customer. This how much a company spends to get each customer. If a company spends R10.00 on advertising to get 100 customers, the spend is 10 cents per customer.

Reference:
Mickency & Company (n.d) Customer lifecycle and analytics insights Available at: https://www.mckinsey.com (Accessed 2 May 2026)

23. CUSTOMER LIFETIME VALUE: A quantity of measure that calculates the total net profit a business yields from a customer over the course of their relationship. The CLV:CAC ratio compares the customer value to the cost of acquiring a customer. If the ratio is 3:1, the business is doing well because the customer is bringing in two times more value.

$$\text{CLV:CAC ration} = \frac{\text{customer Lifetime Value}}{\text{Customer Acquisition Cost}}$$

Reference
Harvard Business Review (n.d.) Customer loyalty and NPS, data-driven strategy Available at: https://hbr.org

24. CHURN RATE: A percentage of customers terminate their ~~business~~ relationship with a business over a specific period of time.

$$\text{Churn Rate} = \frac{\text{Customers Lost}}{\text{Total Customers}} \times 100$$

Reference: Harvard Business Review (n.d.) Customer loyalty and NPS, data-driven strategy Available: hbr.org

25 CUSTOMER RETENTION RATE: A percentage of customers a company retains over a specific period of time.

$$\text{CRR} = \frac{\text{Customers @ end of period} - \text{customers acquired during period}}{\text{customers @ end of period}}$$

Reference: Harvard Business Review (2014) The Value of Keeping the Righ Customers Available at: https://hbr.org

26. Net Promoter Score (NPS): This is a metric that measures the willingness of a company's customer to refer other customers to the company/business.

Harvard Business Review (2003) The One Number You Need to Grow Available at: https://hbr.org/2003

27. AVERAGE ORDER VALUE: This is the average spend of a customer per purchase they make over a certain period of time.

$$AOV = \frac{\text{Total Revenue}}{\text{Number of orders}}$$

HubSpot Academy (n.d.) Sales Funnel, conversion rate, and marketing KPIs Available at: https://academy.hubspot.com

28 SALES FUNNEL: A step-by-step process of a customer having knowledge about a company's products or services until they make a purchase. The funnel stages are Awareness, Interest, Consideration and Conversion, Retention.

Reference: HubSport Academy (n.d.) What is a Marketing Funnel and How Does It Work for Lead Generation? Available at: https//: www.hubspot.com/glossary/marketing-funnel

04 MARKETING & CAMPAIGN ANALYTICS

29 MARKETING CAMPAIGN: An organised set of activities aimed at promoting a product, service, or brand to a defined group of customers. This is mainly the foundation of which the success of a marketing campaign is laid and is the starting point.

Reference: Mckinsey & Company (n.d.) Marketing analytics and attribution. Available at: https://www.mckinsey.com

30. <u>POST - CAMPAING - ANALYSIS</u>: This asseses the success of the campaign's performace using data and to insightful future decisions. Return On Invest, Incremental Revenue, Click-Through-Rate and Return On Ad Spend are use to evaluate success of the campaign.

Reference:
Harvard Business Review (n.d.) Measuring marketing effectiveness and ROI. Available at: https://hbr.org

31. <u>INCREMENTAL REVENUE</u>: This is the extra revenue made as a result of the activities of the marketing campaign over and above the normal sales.

ICR = Campaign week sales - Normal week sales

This is a key input in calculating Return on Invest and Return On Ad Spend

Reference: Harvard Business Review (n.d.) Measuring marketing effectiveness and ROI. Availableat: https://hbr.org

32 <u>RETURN on INVESTMENT</u>: This measures how much profit an investment yield in relation to its cost.

$$ROI = \frac{\text{Icremental Revenue} - \text{CampaignCost}}{\text{Campaign Cost}} \times 100$$

ROI uses Incremental Revenue to evaluate whether the campaign was worth it.

Reference: Investopedia (2026) What is Return on Investment (ROI) and How to Calculate it. Availableat: https://www.investopedia.com/terms/r/returnoninvestment.asp

33. Return on Ad Spend (ROAS): This calculates the revenue gained relative to each unit of currency (rand) spent on advertising. It's the ratio of revenue gained from advertising to advertising spend

$ROAS = \frac{Revenue}{Ad\ Spend}$ Spend: R2000 Revenue: R10000 ROAS: 5:1

Here ROAS : focuses on revenue
ROI : focuses on profit

Reference
Google Analytics Help (n.d) CTR, bounce rate, attribution, ROAS. Available at: https://support.google.com/analytics

34. Clik-Through Rate (CTR): This rate calculates the percentage of prospective customers who see a company's advert and click on it.

$CTR = \frac{\text{Number of Clicks}}{\text{Number of Impressions}} \times 100$ ∴ If 500 prospective customers see your advert and 50 click, CTR = 10%

CTR relates to the SALES FUNNEL in that if it is low, it indicates that your marketing strategy does not align with whomever you are targeting it to because the prospective customer is not going beyond the AWARENESS stage until they take action to make a purchase (conversion). Therefore you are spending IMPRESSIONS that don't turn into action

Reference:
Google Analytics Help (n.d.) CTR, bounce rate, attribution, ROAS. Available at https://support.google.com/analytics

35. BOUNCE RATE: This is a percentage of visitors on a website who leave it not having taken action beyond just viewing it. It measures the first interaction not conversion. It indicates whether your page is able to maintain att tion within the first few seconds

$$\text{Bounce Rate} = \frac{\text{Single-Page Sessions}}{\text{Total Sessions}} \times 100$$

↑ high bounce rate: page does not resonate with visitors

↓ low bounce rate: visitors are interaction with your site beyond the landing page

Referecence:

Google Analytics Help (2024) Bounce Rate Definition Available at: support.google.com (Accessed: 3 May 2026)

36. A/B TESTING: When you have two variations of a webpage, advert or product feature that you test on a specific audience using metrics such as conversion rate, average order value or click-throug rate to compare which one performs better. This method boosts incremental revenue and Return on Ad Spend without increasing spend.

$$\text{Lift} = \frac{\text{Metric}_{variant} - \text{Metric}_{control}}{\text{Metric}_{control}} \times 100$$

control = 1^{ST} webpage version

variant = 2^{nd} webpage version

Advert A = 50%

Advert B = Buy 1 Get 1 Free

Reference

Meta Platforms, Inc (2026) WhatsApp Meta AI [Large Language Model] Available at: WhatsApp (Accessed: 3 May 2026)

37. LEAD GENERATION: This process entails drawing the attention of prospective customers compiling their information, with the aim of nverting the inteaction into a sal . This is for potential customers indicating interest in the offering-

This relates to the stage in the SALES FUNNEL of AWARENESS of capturing attention through social media and other content. (see SEO)

$$\text{COST PER LEAD (CPL)} = \frac{\text{Campaign Spend}}{\text{No. of Leads Generated}}$$

Reference:
HubSpot Academy (n.d.) Marketing campaigns, A/B testing, and lead generation. Available at: https://academy.hubspot.com

38. MARKETING ATTRIBUTION: This is a process of assigning or allocating credit to the point of contact responsible for the conversion or the sale made. This is essential for budget allocation and understanding Return On Investment. A customer doesnt always buy instantly. They might make a conversion later on after going through various marketing channels. They may see a Facebook post then later search in Google and make a conversion.

Reference:
Google Analytics Help (n.d.) CTR, bounce rate, attribution, ROAS. Available at: https://support.google.com/analytics